CONNECTING & SHAPING DATA

FRONT-END VS. BACK-END

Power BI Desktop essentially has two distinct environments: a **front-end and a back-end**

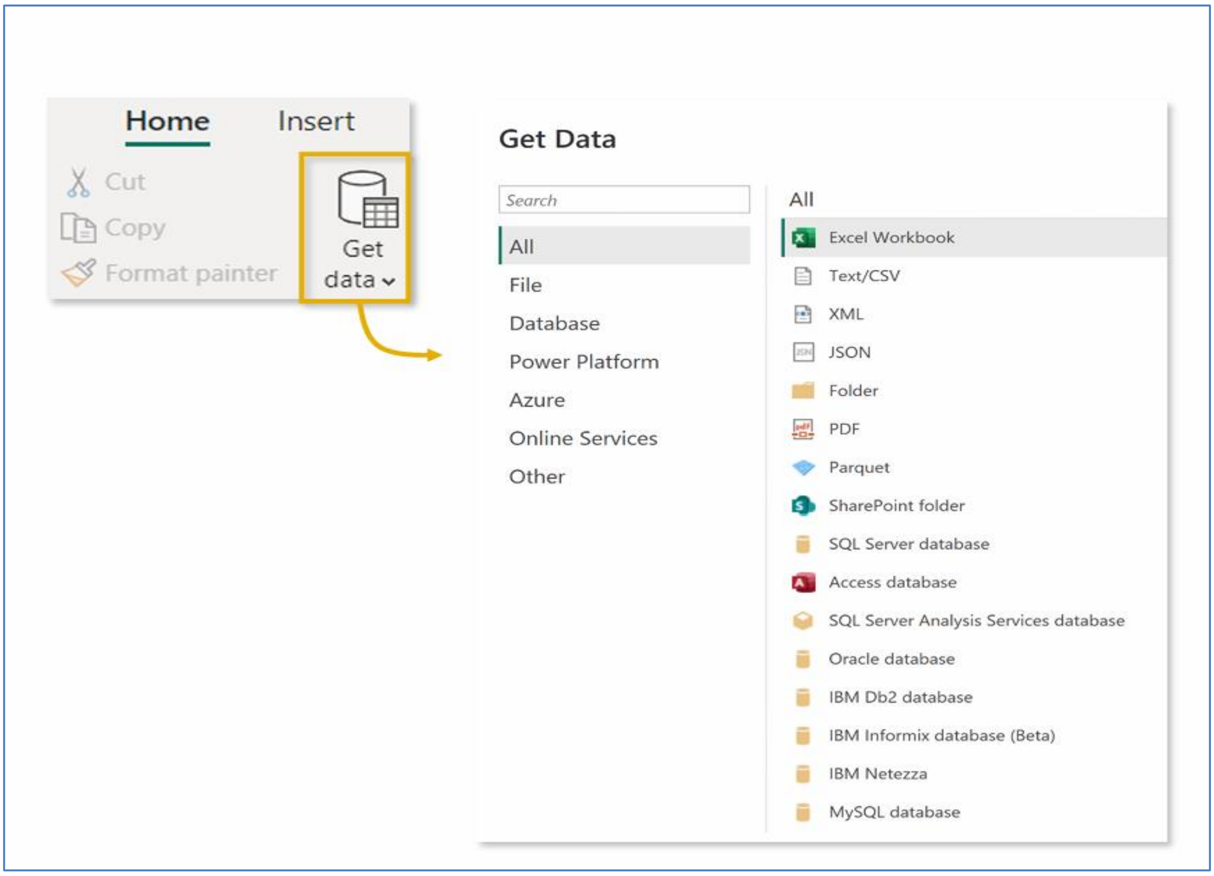
• The **front-end** includes the **Data, Model & Report** views, where most of the modeling, analysis and visualization takes place

• The **back-end** includes the **Power Query Editor**, where raw data is extracted, transformed, and loaded to the front-end (ETL)

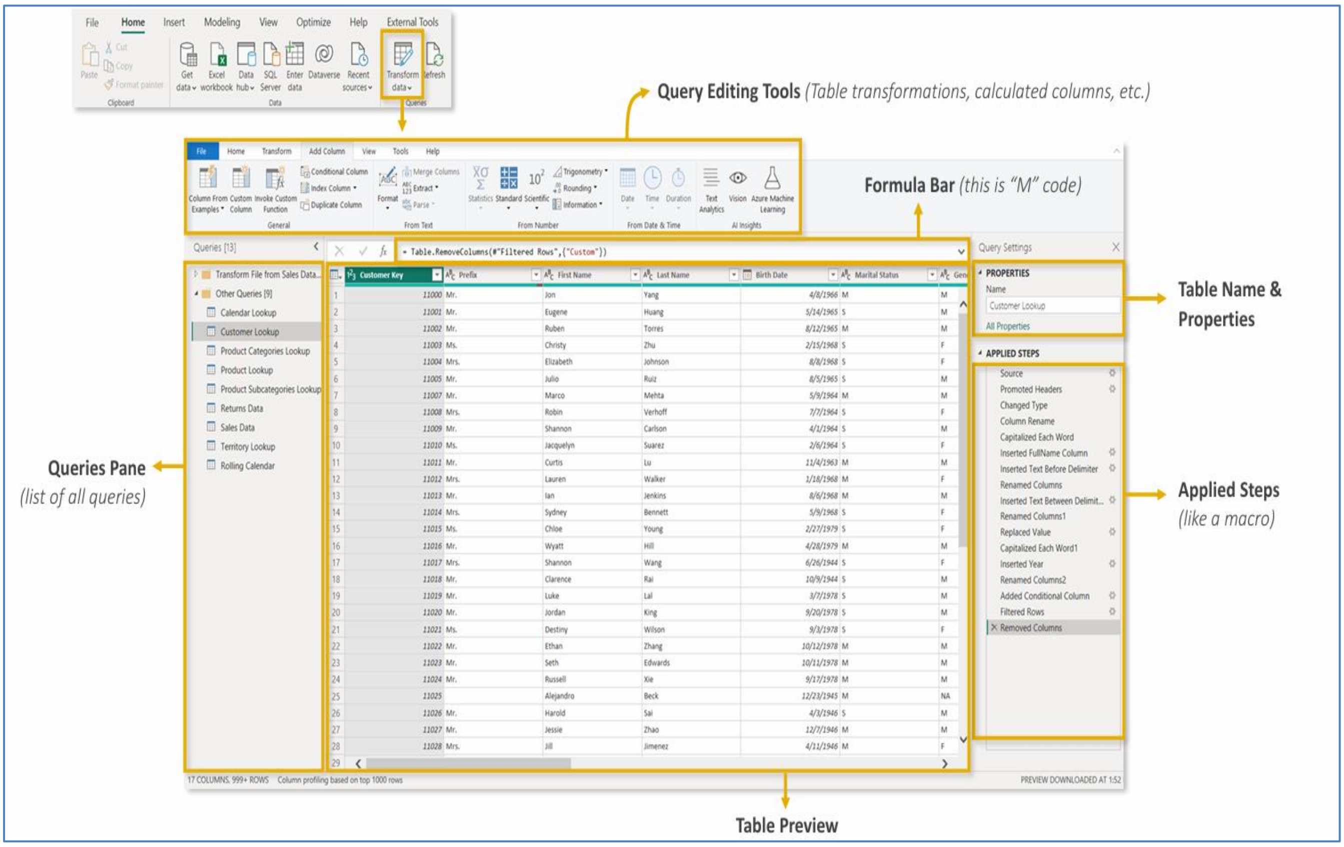
TYPES OF DATA CONNECTORS

Power BI can connect to virtually any type of source data, including (but not limited to):

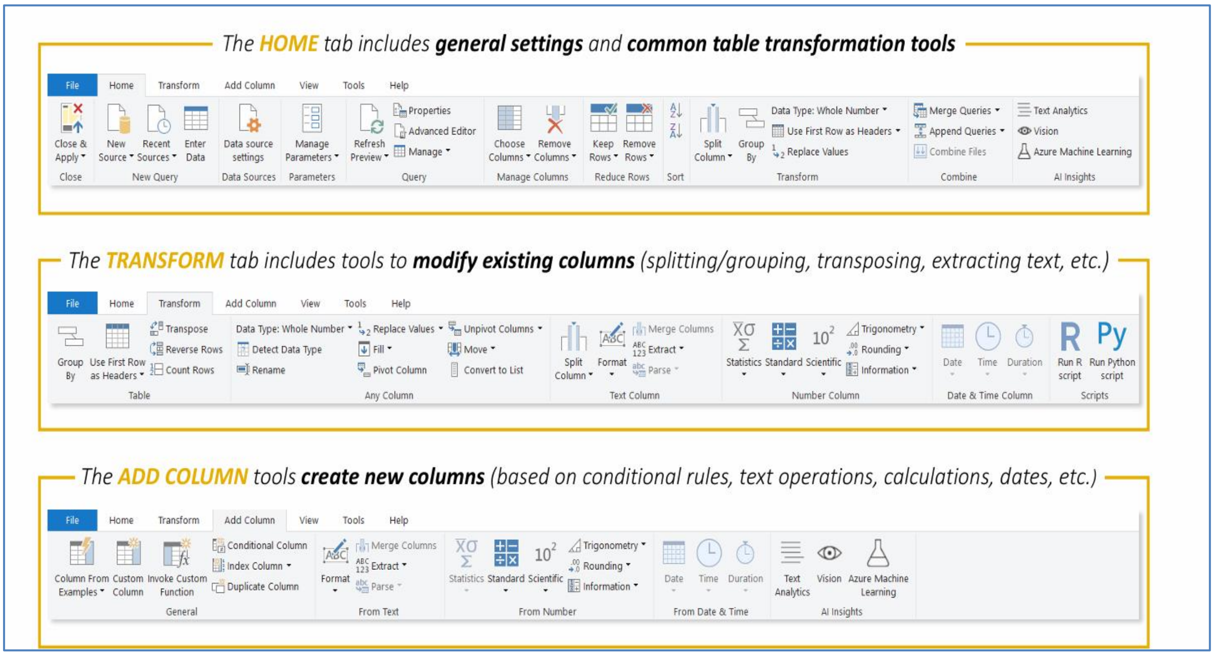
- Flat files & Folders (csv, text, xlsx, etc.)
- Databases (SQL, Access, Oracle, IBM, etc.)
- Power Platform (Datasets, DataMart's, Dataflows, Dataverse, etc.)
- Azure (Azure SQL, Analysis Services, Databricks, etc.)
- Online Services (SharePoint, GitHub, Dynamics 365, Google Analytics, Salesforce, Power BI Service, etc.)
- Other (Web feeds, R scripts, Spark, Hadoop, etc.)



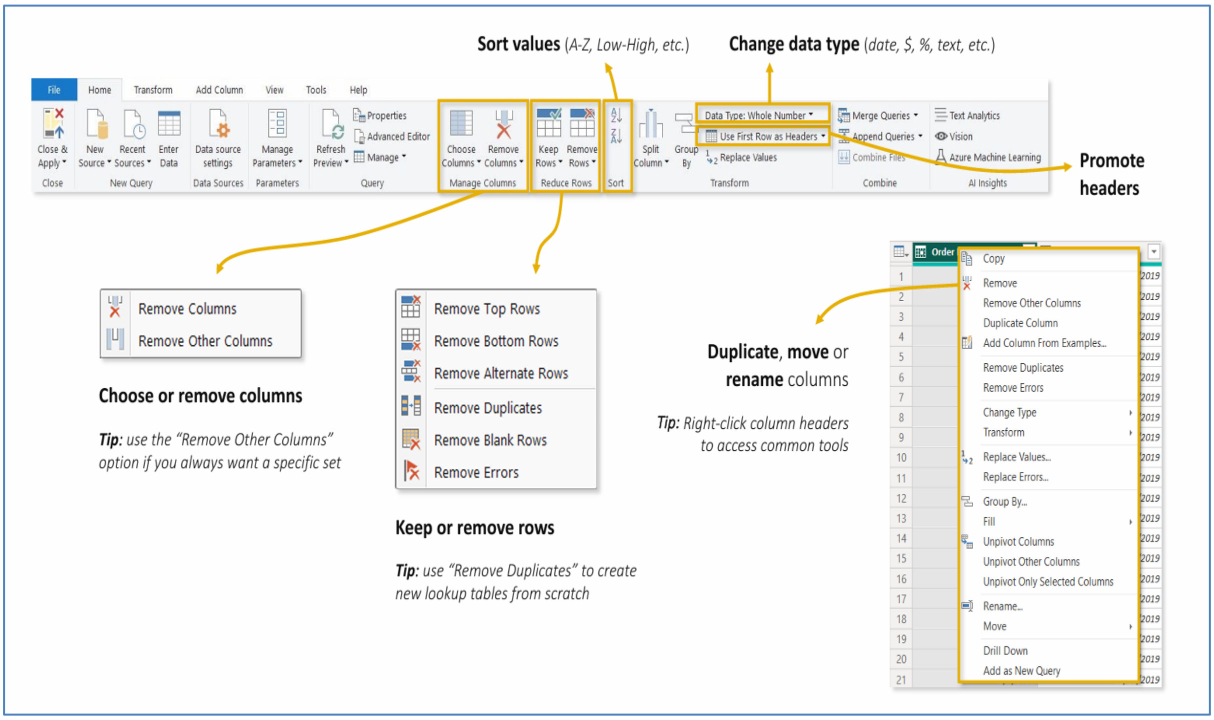
POWER QUERY EDITOR



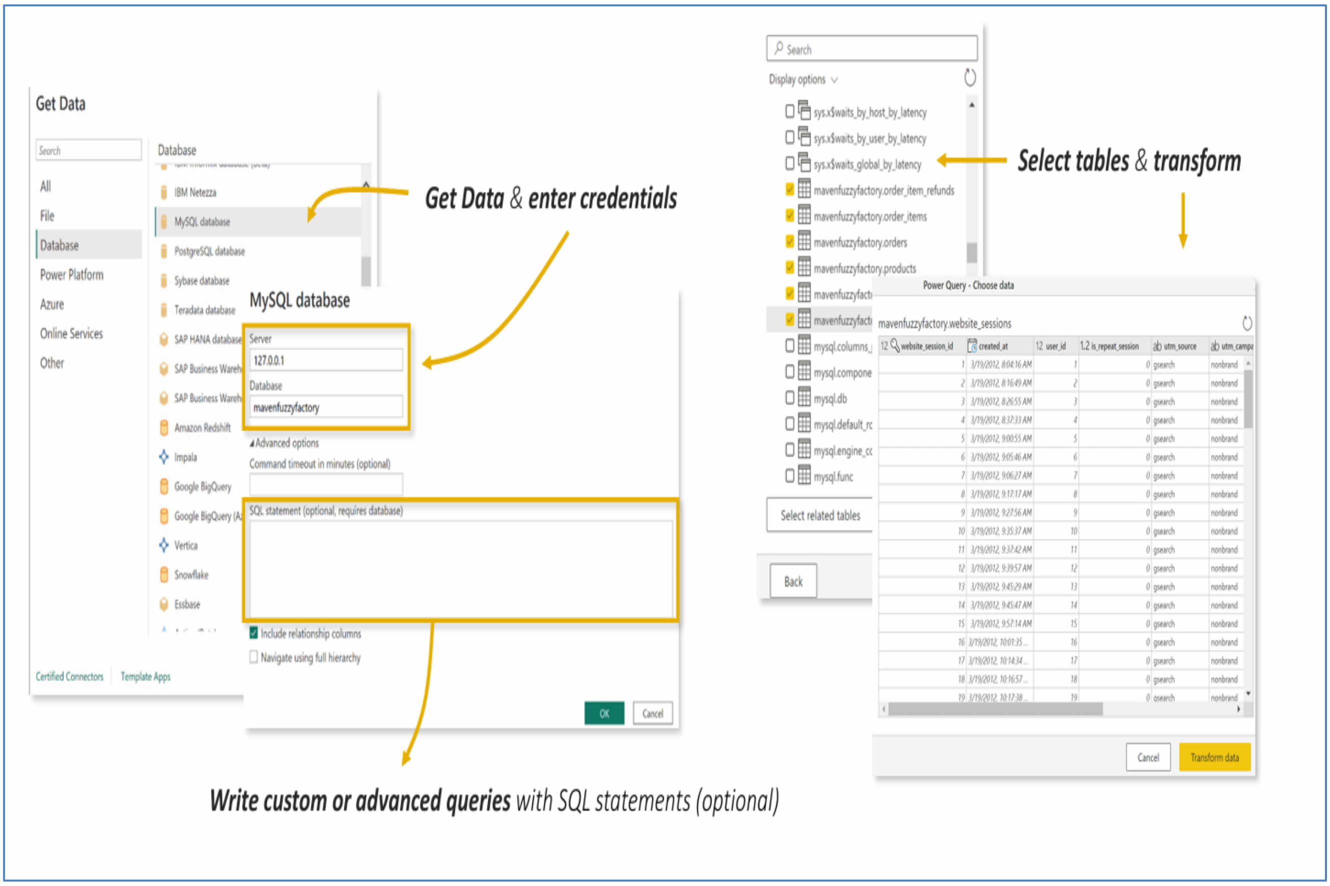
QUERY EDITING TOOLS



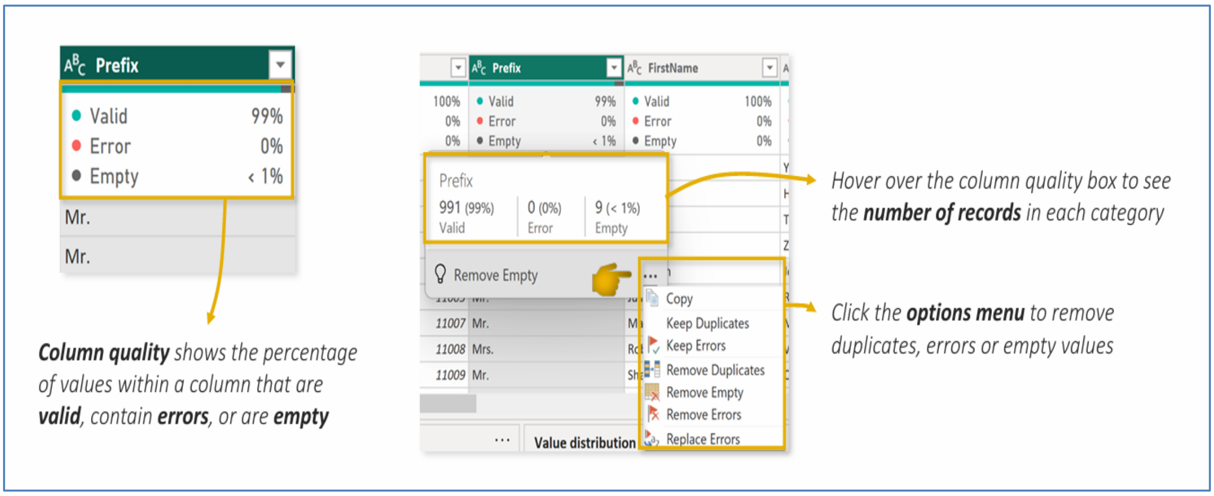
BASIC TABLE TRANSFORMATIONS



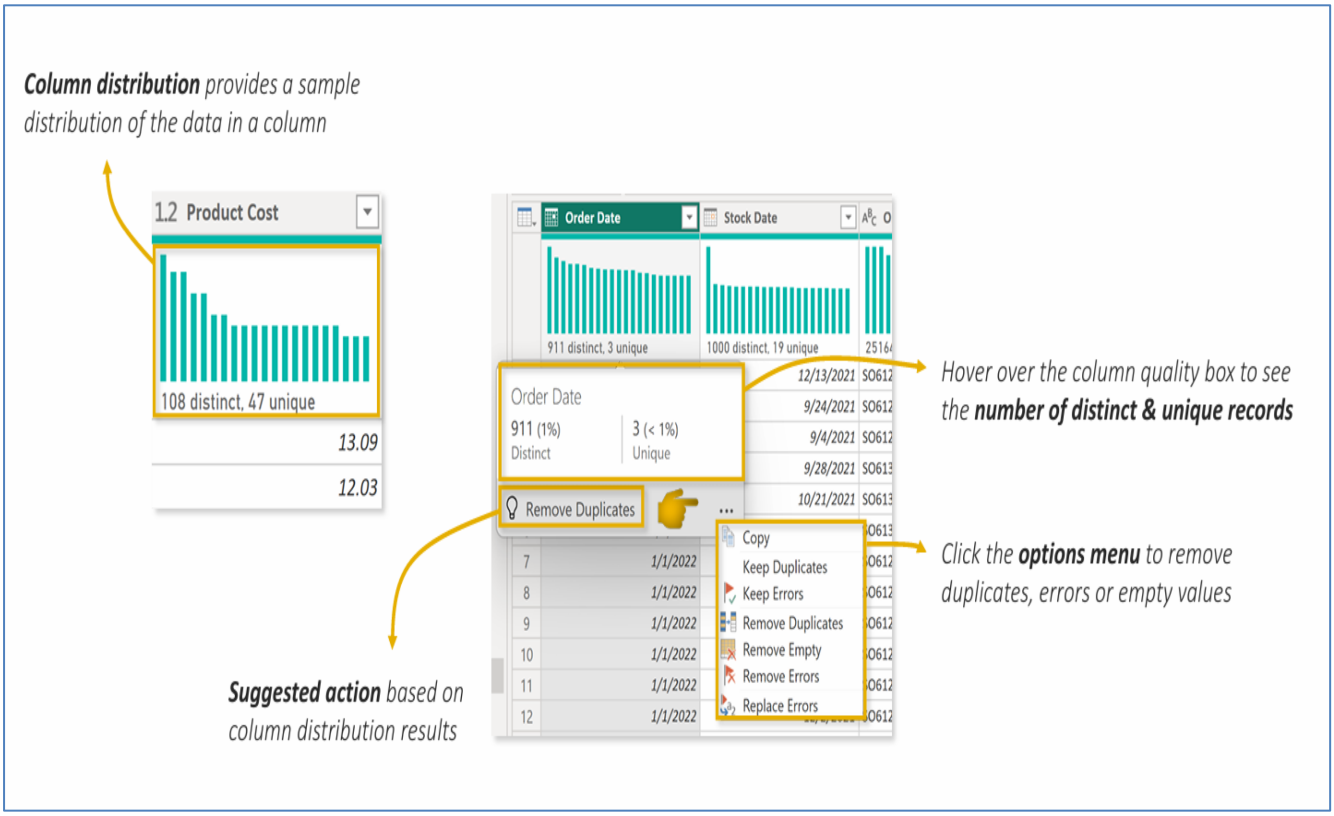
CONNECTING TO A DATABASE



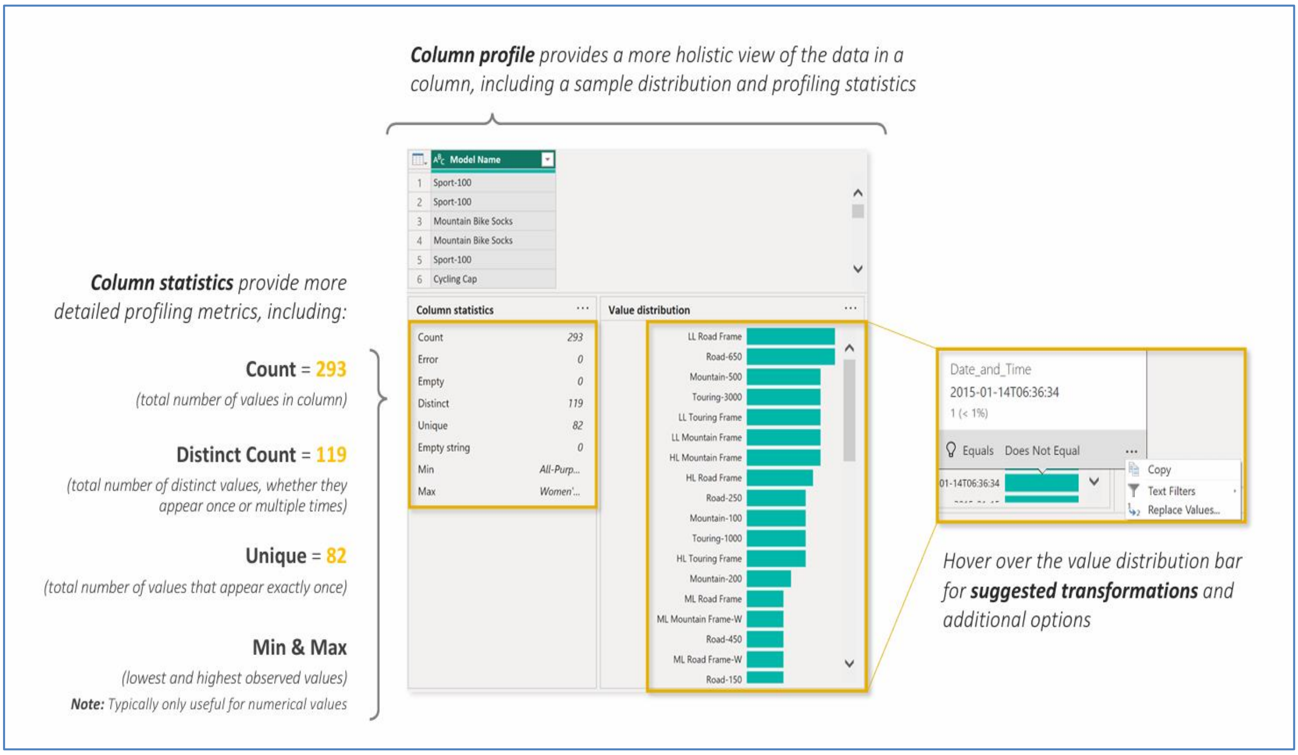
DATA PROFILING: COLUMN QUALITY



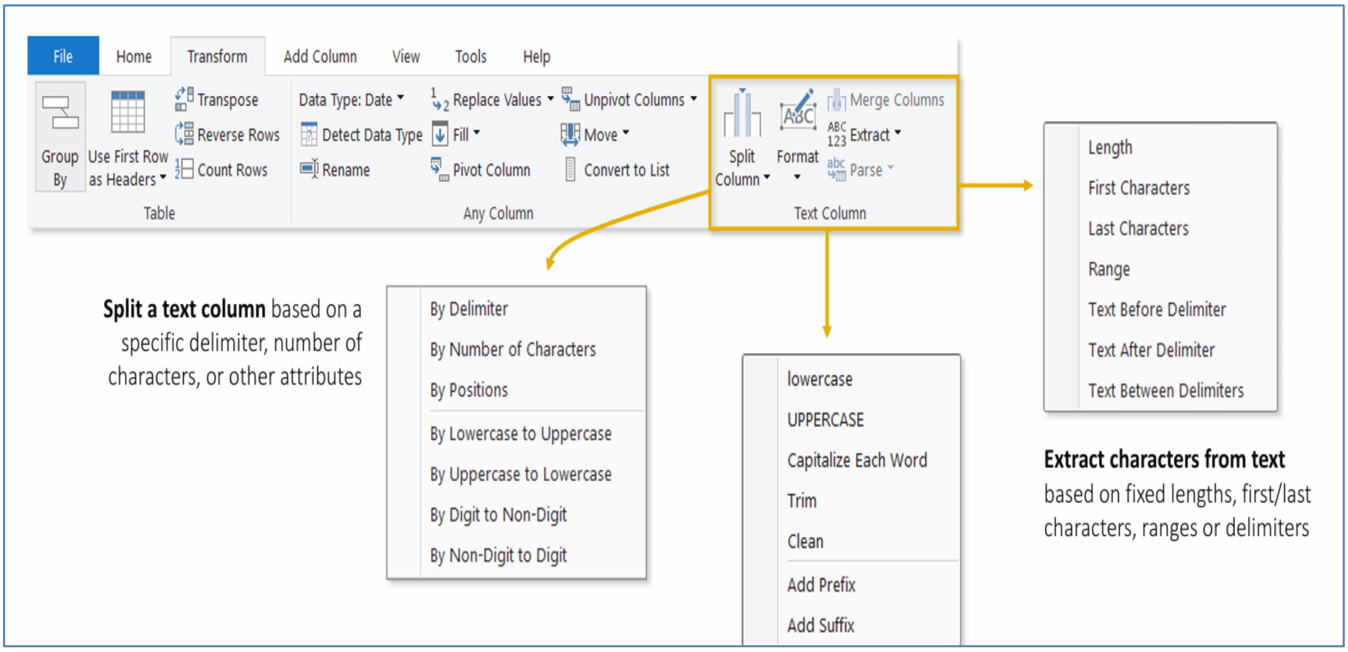
DATA PROFILING: COLUMN DISTRIBUTION



DATA PROFILING: COLUMN PROFILE

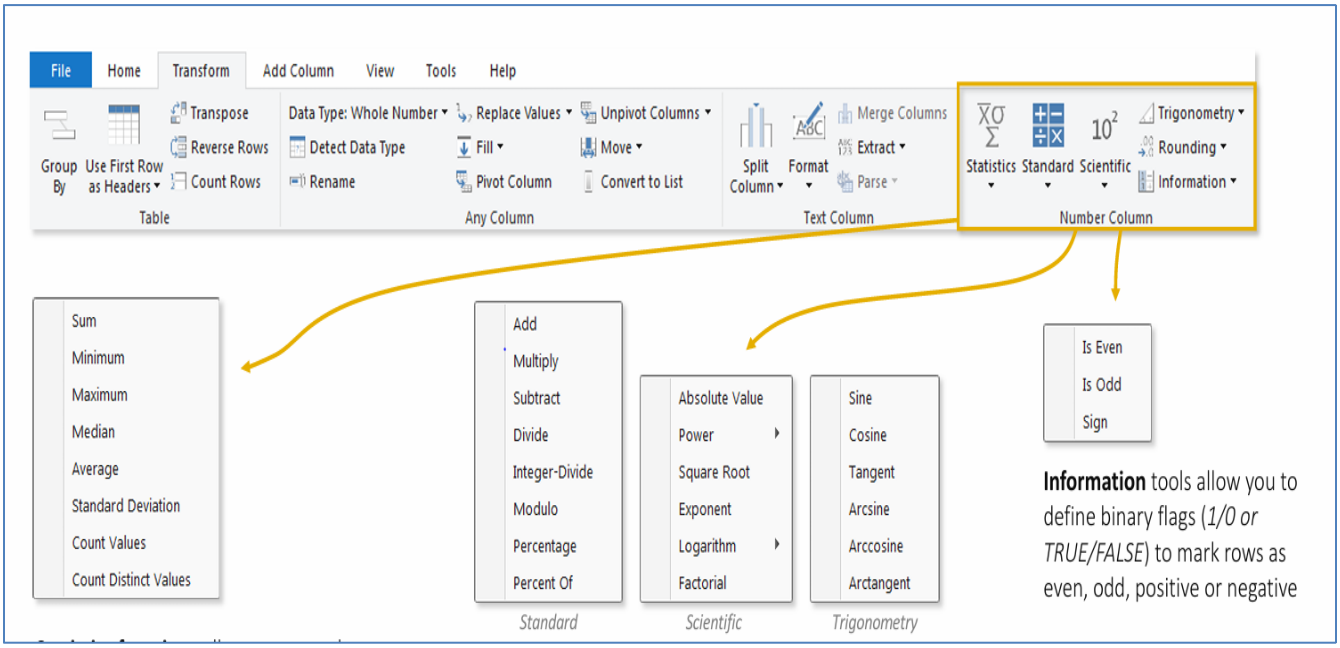


TEXT TOOLS



You can access many tools from both the **Transform** and **Add Column** menus - the difference is whether you want to ADD a new column or **OVERWRITE** an existing one.

<u>NUMERICAL TOOLS:</u>



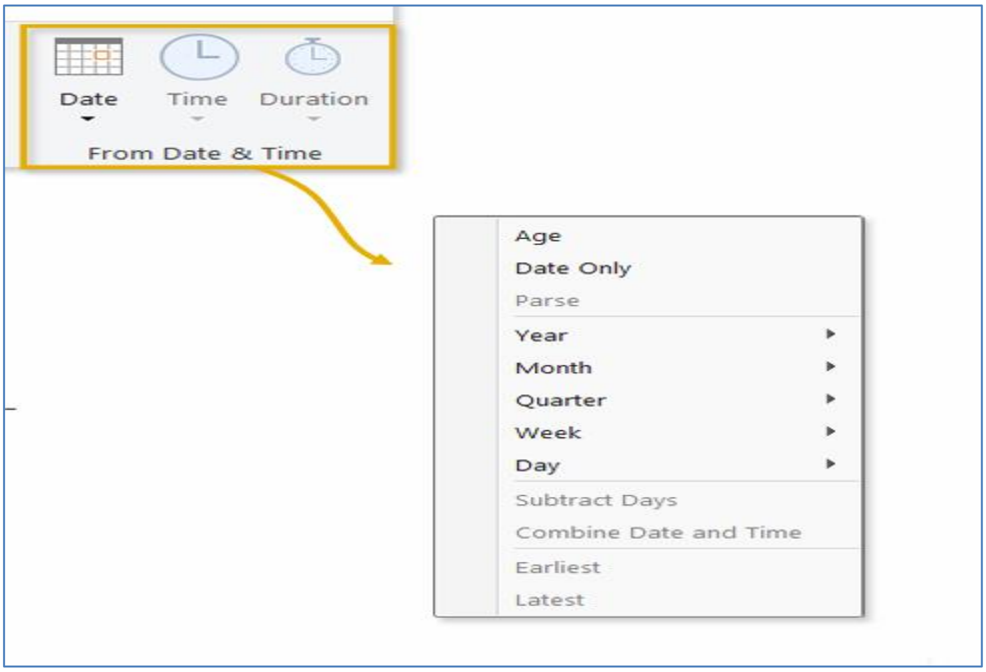
Statistics functions allow you to evaluate basic stats for a selected column (sum, min/max, average, count, count distinct, etc.)

Standard, Scientific and **Trigonometry** tools allow you to apply standard operations (addition, multiplication, division, etc.) or more advanced calculations (power, logarithm, sine, tangent, etc.) to each value in a column

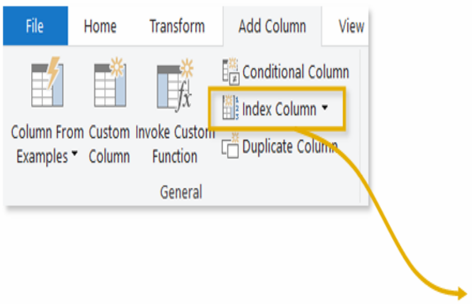
<u>DATE & TIME TOOLS</u>

Date & Timetools are relatively straight-forward, and include the following options:

- Age: Difference between the current date and the date in each row
- Date Only:Removes the time component from a date/time field
- Year/Month/Quarter/Week/Day: Extracts individual components from a date field (time specific options include Hour, Minute, Second, etc.)
- Earliest/Latest: Evaluates the earliest or latest date from a column as a single value (can only be accessed from the “Transform” menu)



INDEX COLUMNS

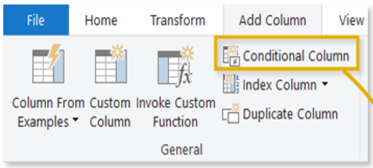


Index Columns contain a list of sequential values that can be used to identify each unique row in a table (*typically starting from 0 or 1*)

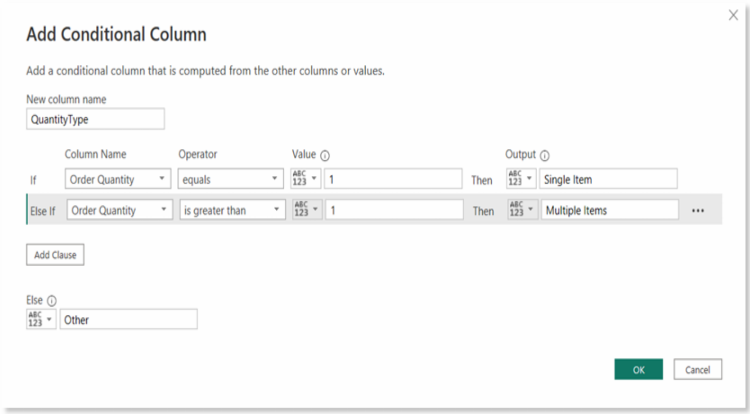
These are often used to create **unique IDs** that can be used to form relationships between tables (*more on that later!*)

	Index	Order Date	Stock Date	Order Number	Product Key
1	1	1/1/2020	9/21/2019	SO45080	332
2	2	1/1/2020	12/5/2019	SO45079	312
3	3	1/1/2020	10/29/2019	SO45082	350
4	4	1/1/2020	11/16/2019	SO45081	338
5	5	1/2/2020	12/15/2019	SO45083	312
6	6	1/2/2020	10/12/2019	SO45084	310
7	7	1/2/2020	12/18/2019	SO45086	314
8	8	1/2/2020	10/9/2019	SO45085	312
9	9	1/3/2020	10/3/2019	SO45093	312
10	10	1/3/2020	9/29/2019	SO45090	310
11	11	1/3/2020	12/11/2019	SO45088	345
12	12	1/3/2020	10/24/2019	SO45092	313
13	13	1/3/2020	12/16/2019	SO45089	351
14	14	1/3/2020	10/26/2019	SO45091	314
15	15	1/3/2020	9/11/2019	SO45087	350
16	16	1/3/2020	9/11/2019	SO45094	310
17	17	1/4/2020	10/30/2019	SO45096	312
18	18	1/4/2020	10/30/2019	SO45097	313
19	19	1/4/2020	9/15/2019	SO45098	310
20	20	1/4/2020	12/7/2019	SO45095	344

CONDITIONAL COLUMNS

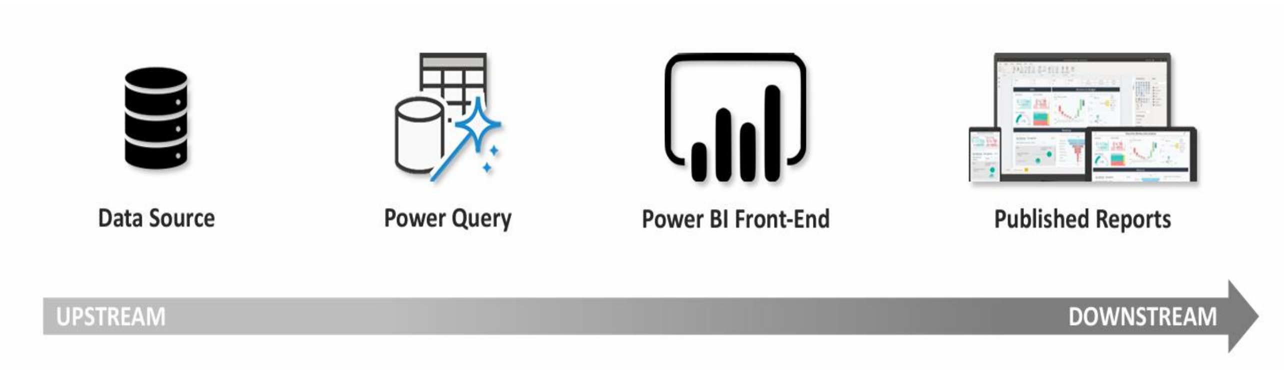


Conditional Columns allow you to define new fields based on logical rules and conditions (IF/THEN statements)

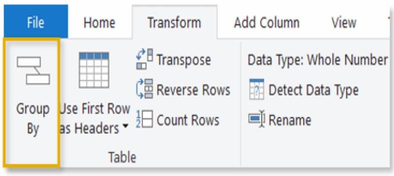


Here we're creating a conditional column named **Quantity Type**, which is based on **Order Quantity**:

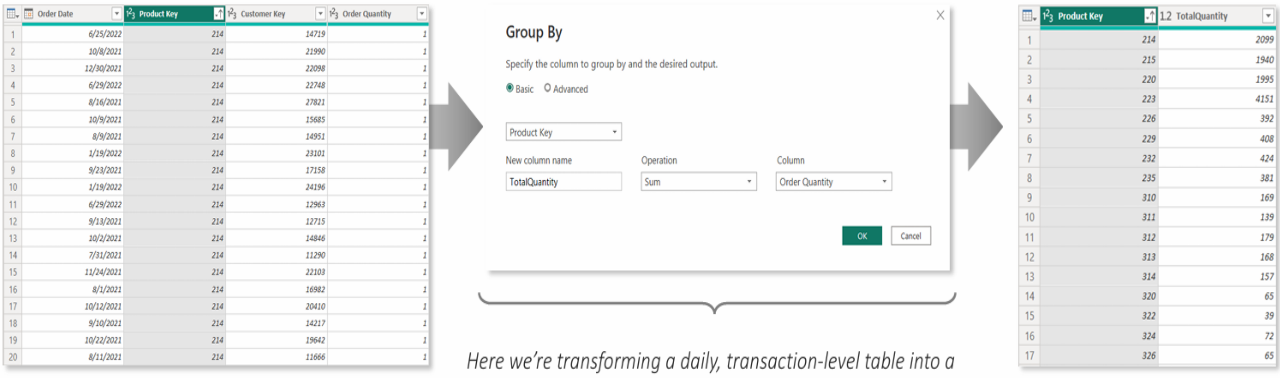
- *If Order Quantity* ***=1****, Quantity Type = "****Single Item****"*
- *Else If Order Quantity* ***>1****, Quantity Type = "****Multiple Items****"*
- *Else; Quantity Type = "****Other****"*



GROUPING & AGGREGATING

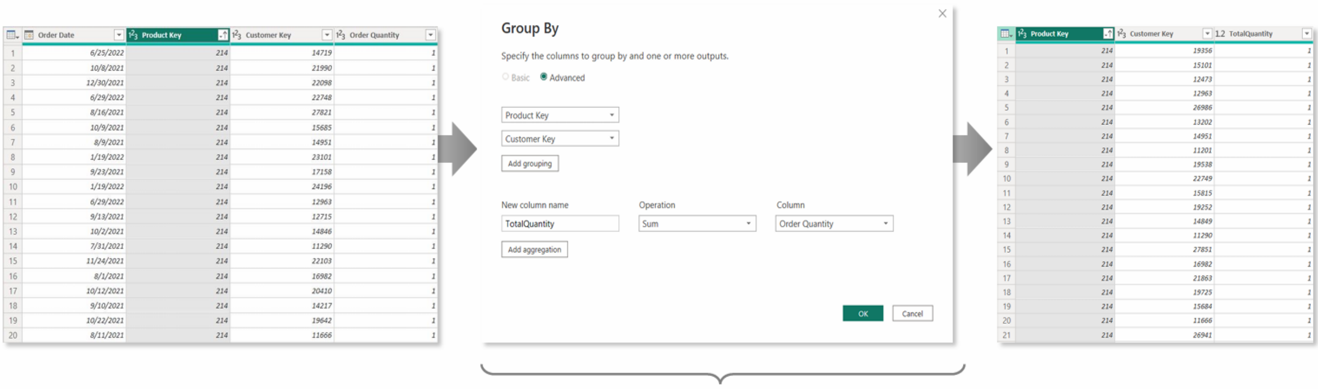


Group By allows you to aggregate data at a different level or "grain"
(i.e. group daily records into monthly, aggregate transactions by store, etc.)



*Here we're transforming a daily, transaction-level table into a summary of **Total Quantity** by **Product Key***

***NOTE:** Any fields not specified in the Group By settings are lost*

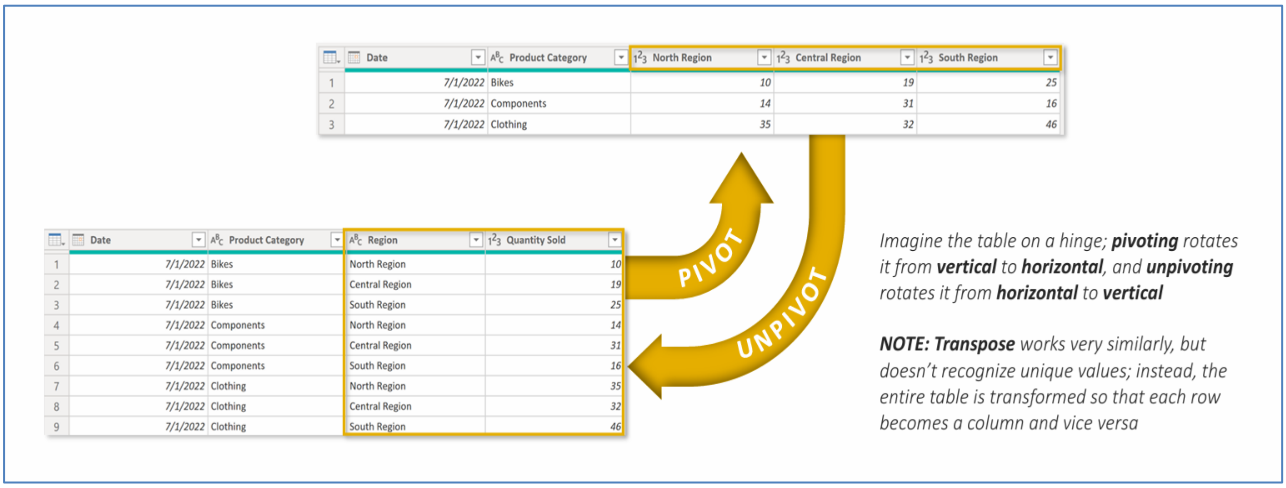


*This time we're transforming the daily, transaction-level table into a summary of **Total Quantity** grouped by both **Product Key** and **Customer Key** (using the "Advanced" option)*

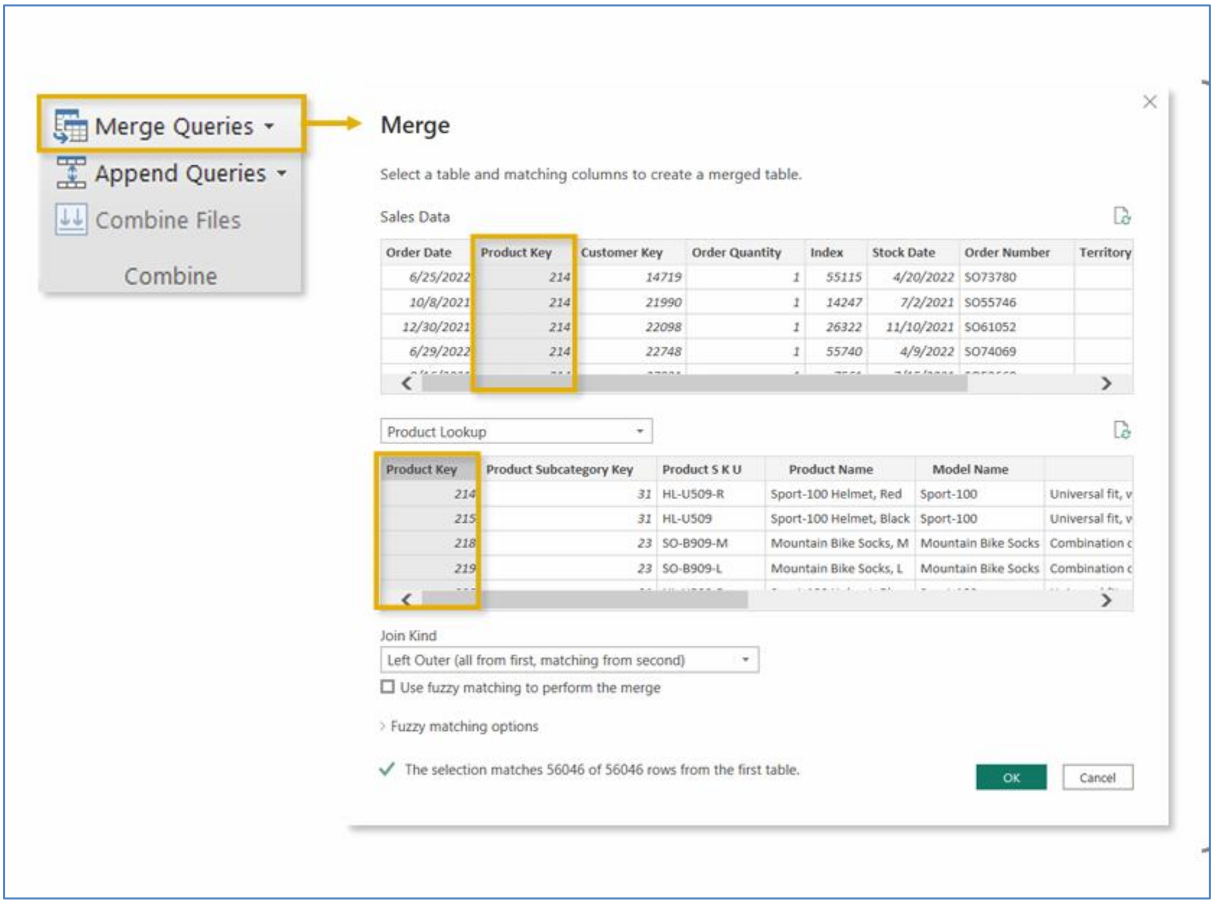
***NOTE:** This is like creating a PivotTable in Excel and pulling in **Sum of Order Quantity** with **Product Key** and **Customer Key** as row labels*

PIVOTING & UNPIVOTING

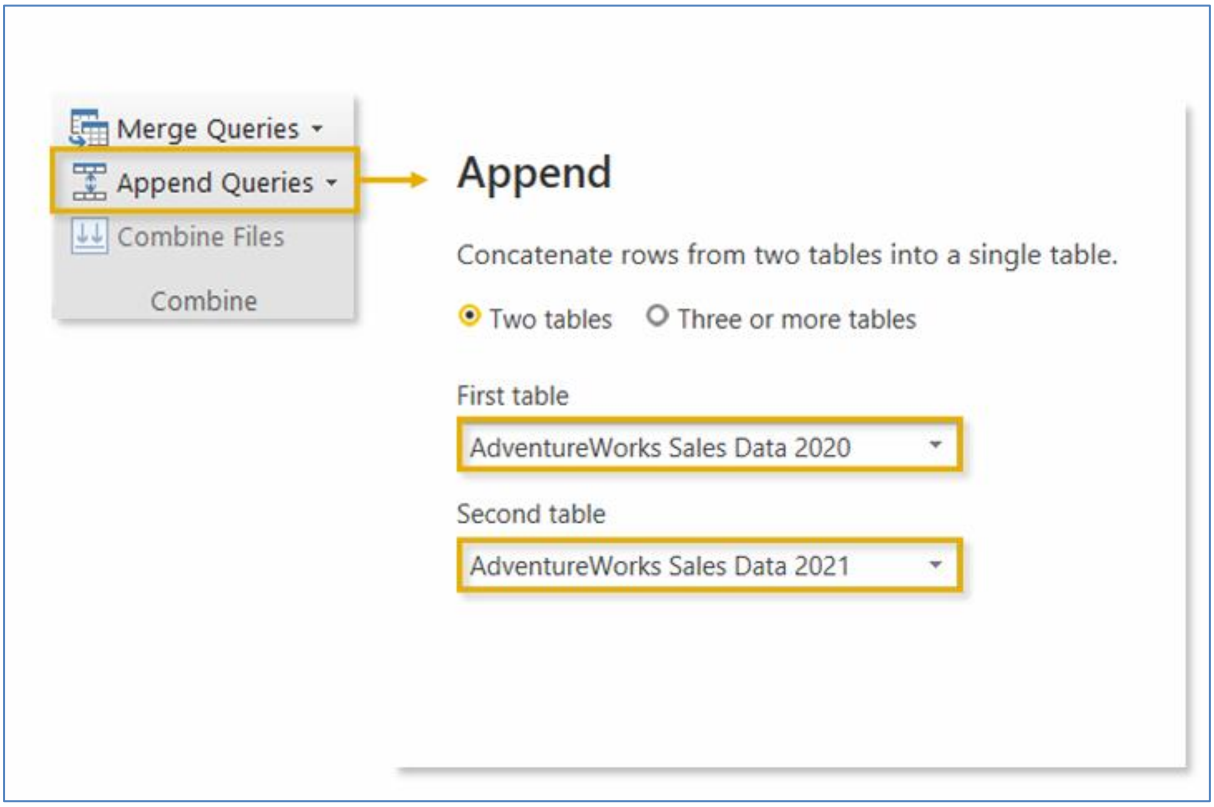
Pivoting describes the process of turning distinct row values into columns, and unpivoting describes the process of turning distinct columns into rows



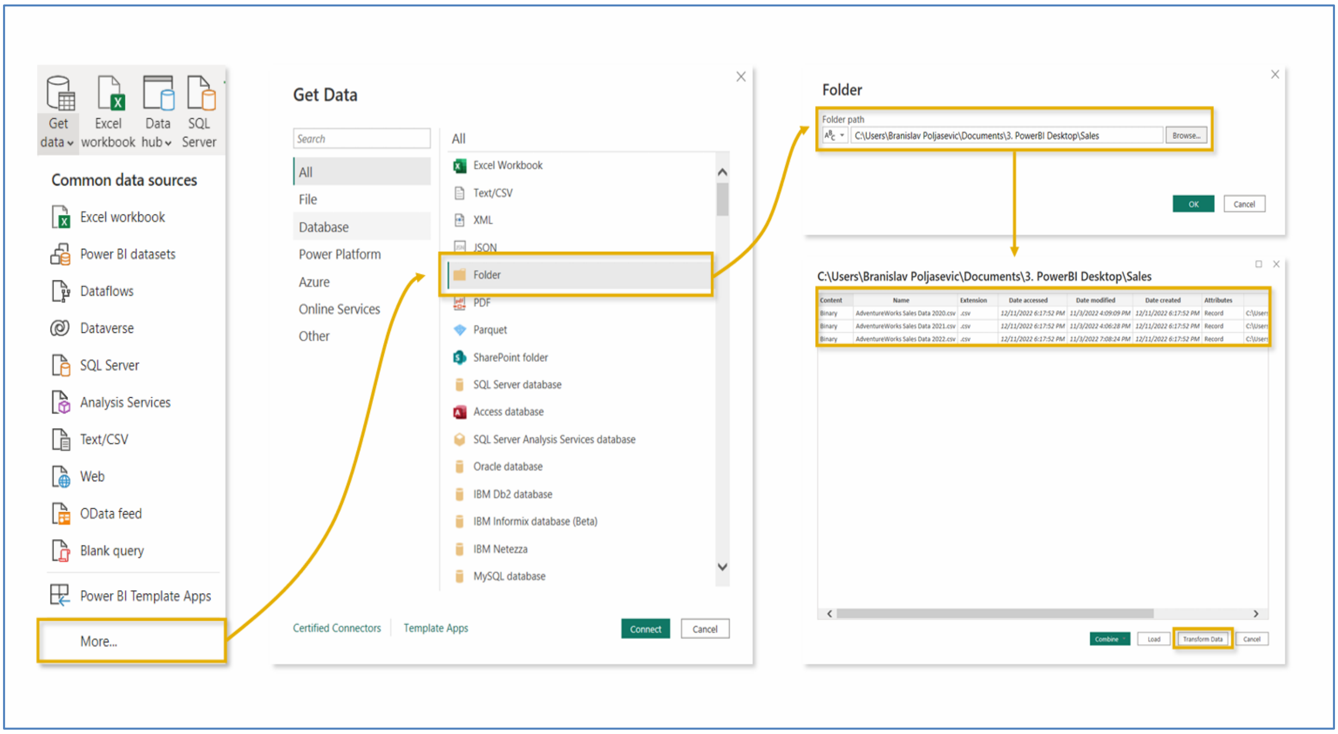
MERGING QUERIES



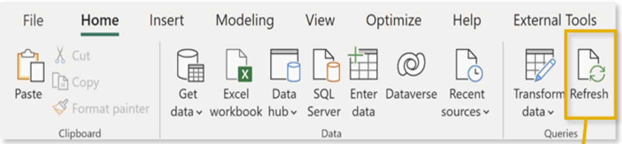
APPENDING QUERIES



APPENDING FILES FROM A FOLDER

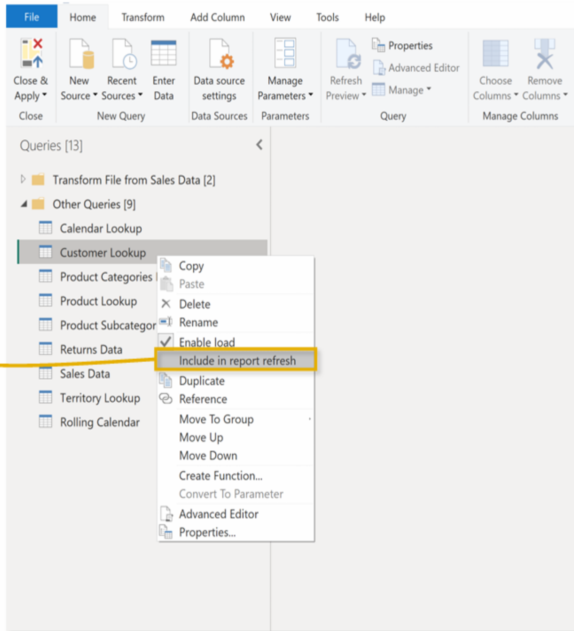


<u>REFRESHING QUERIES</u>



By default, ***all queries*** will refresh when you use the **Refresh** command from the **Home** tab

From the Query Editor, uncheck **Include in report refresh** to exclude individual queries from the refresh



PRO TIP: Exclude queries from refresh that don't change often (like lookups or static data tables)